HOME CREDIT DEFAULT ASSESSMENT REPORT

Joseph Aragbaiye

Student ID: 5025912

MSc Big Data Analytics

Sheffield Hallam University

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Overview

This report presents an end-to-end credit default prediction system built on the Home Credit data. The work began with 307,511 labelled applicants and an observed default rate of 8.07%, making class imbalance a central part of the modelling problem. A stratified 80/20 split created 246,008 development cases and 61,503 held-out cases. The test split was kept separate while features, models, calibration and the operating threshold were developed.

The analysis combined application variables with aggregated credit-history information from bureau, bureau balance, previous applications, instalment payments and POS/CASH records. Feature selection used three complementary signals: L1 logistic regression, LightGBM gain importance and permutation importance. Their combined top-ranked features produced a working set of 21 source variables and 98 transformed model columns.

Across a shared five-fold cross-validation design, tuned LightGBM gave the strongest development ROC-AUC at 0.7621, followed closely by XGBoost at 0.7618. Keras and PyTorch implementations of the same multilayer perceptron both achieved about 0.7523, while the strongest classical models were around 0.7466. The Friedman test found evidence of differences across models, although five folds were not sufficient for Bonferroni-corrected pairwise Wilcoxon significance.

The selected LightGBM model required probability calibration because class weighting produced probabilities that were too high. Platt scaling reduced held-out ECE from 0.3167 to 0.0031 without changing ROC-AUC. Using a stated 5:1 false-negative to false-positive cost assumption, an out-of-fold threshold of 0.1613 reduced test cost by 4,277 units compared with 0.50. The final recommendation is therefore a Platt-calibrated LightGBM model used as monitored decision support rather than an automatic lending decision.

1. Introduction

Credit default prediction is not only a classification task. In lending, a model must rank risk reliably, produce probabilities that can be interpreted sensibly, operate at a defensible decision threshold and remain understandable enough to support governance. These requirements are more important when the outcome is imbalanced and decisions affect applicants directly.

The aim of this assessment was to build and compare a broad set of machine-learning approaches on one coherent Home Credit workflow. The study covers data preparation, relational feature engineering, feature reduction, classical models, ensembles, neural networks, statistical comparison, calibration, subgroup analysis, imbalance treatment, explainability and deployment readiness. ROC-AUC was used as the main ranking measure, supported by PR-AUC, recall and Brier score. Accuracy was deliberately not treated as a primary metric because predicting “no default” for every applicant would already be 91.93% accurate while identifying no defaulters.

A further objective was to keep the modelling process connected to practical use. The notebook therefore did not stop at the model with the highest validation score. It also examined whether predicted probabilities were trustworthy, whether performance changed across applicant groups, how imbalance strategies affected results, and whether the final model could be saved, reloaded and scored efficiently. This wider view matters because a small improvement in discrimination has limited value if the resulting system is poorly calibrated, difficult to explain or expensive to operate.

2. Data Preparation and Feature Engineering

The original application table contained 307,511 rows and 122 columns. Defaults accounted for 8.07% of cases, equivalent to an imbalance of about 11.4 non-defaults for every default. A stratified split preserved that rate in both the development and held-out sets. All exploratory and model-development decisions were based on the development data so that the test sample remained a genuine final check.

Missing data was substantial. Sixty-seven of the 122 application columns contained missing values, with more than 7.3 million missing cells in the development split. Some property and housing variables were absent for more than half of applicants. Missingness was therefore treated as a feature of the dataset rather than a minor cleaning problem. The notebook also checked whether missingness could plausibly depend on observed information. For example, OWN_CAR_AGE was considered structurally missing when an applicant reported

no car, while EXT_SOURCE_1 and EXT_SOURCE_3 were treated as carrying possible MNAR risk because availability may depend partly on external information that is not observed in the application table.

Domain anomalies were handled separately from ordinary NaNs. A notable example was the DAYS_EMPLOYED sentinel value of 365243, which is not a realistic employment duration. Instead of discarding the information completely, the workflow created an anomaly indicator and treated the implausible duration appropriately during feature preparation.

Applicant-level features were then created to express relationships that raw columns do not show directly. These included affordability ratios, the gap between credit and goods price, social-circle default summaries, bureau enquiry summaries and income per family member. The relational part of the dataset was handled with a strict aggregate-before-join rule. Bureau balance was first reduced to bureau-record level; bureau, previous applications, instalment payments and POS/CASH histories were then aggregated to one row per applicant before they were left-joined to the application spine. Row-count assertions confirmed that the joins did not duplicate applicants. Forty-one relational features were documented, and explicit indicators were added for applicants with no history in a particular source.

After engineering and joining, the modelling frame contained 173 candidate input columns before one-hot expansion. This stage was important because it converted separate transaction and history tables into applicant-level evidence without leaking repeated records into the modelling sample.

Figure 1 makes the imbalance problem visible. The non-default class dominates the training sample, while defaults form only a small minority. The baseline comparison also shows why accuracy would give a misleading impression of success. A classifier that always predicted the majority class could achieve high accuracy, yet it would have no practical ability to identify risky applicants. ROC-AUC and PR-AUC baselines therefore provide more meaningful reference points for later models. The missingness and anomaly checks in **Figure 2** also support preserving informative absence indicators and treating unusual values carefully rather than applying one cleaning rule to every variable.

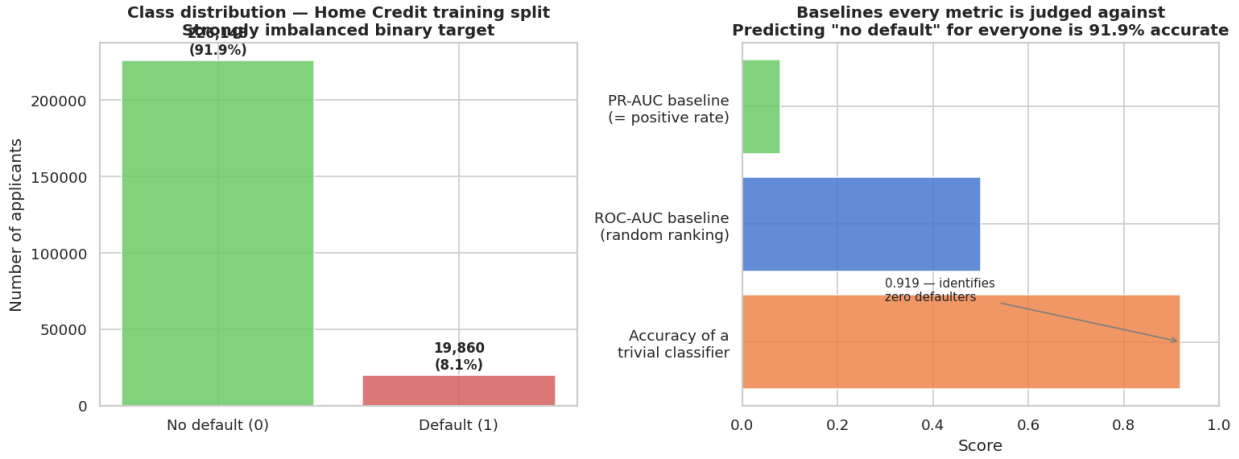


Figure 1. Training class distribution and no-skill baselines. Defaults account for only 8.07% of applicants, showing why accuracy alone is unsuitable.

Source: executed output from the submitted Home Credit assessment notebook.

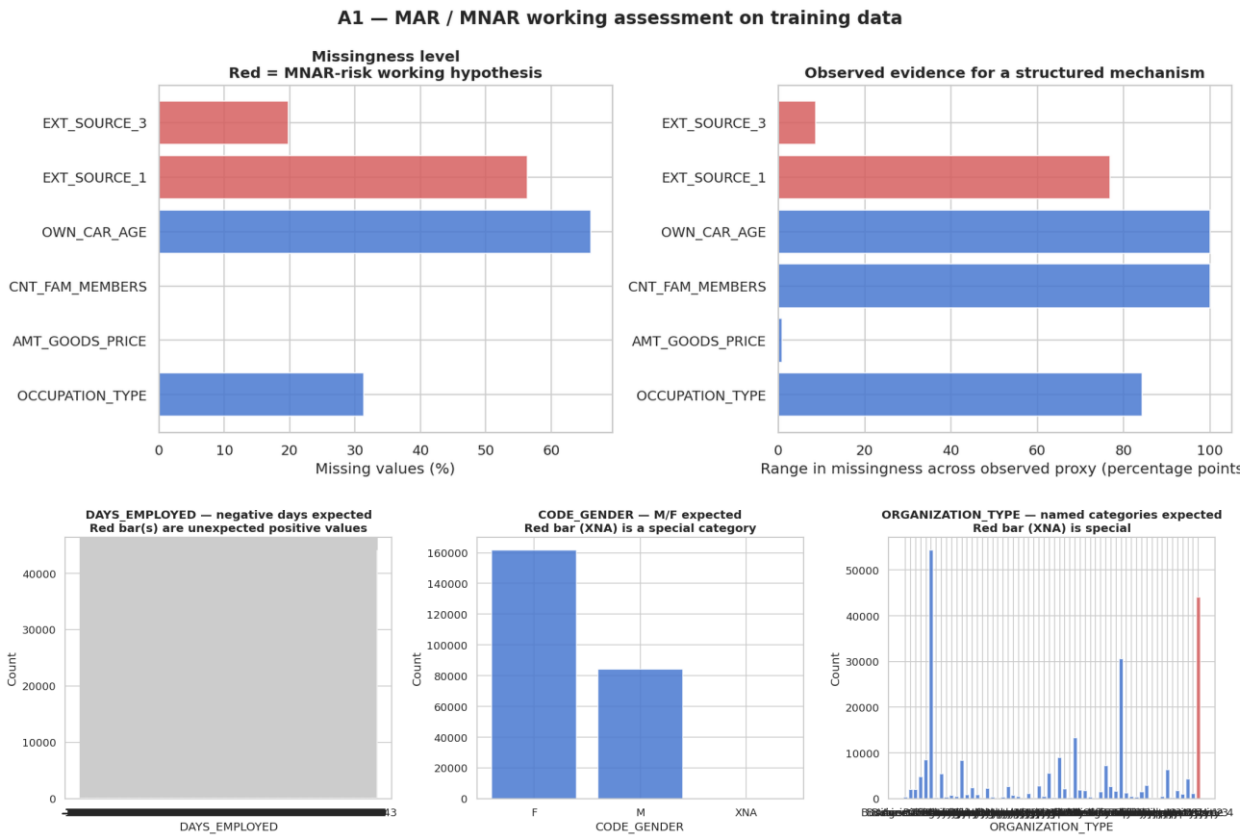


Figure 2. Missingness working assessment and domain-anomaly diagnostics. The notebook separates structured missingness from implausible sentinel values before modelling.

Source: executed output from the submitted Home Credit assessment notebook.

3. Feature Selection and Preprocessing

A crude preprocessing matrix was first created for screening. Median imputation and scaling were applied to numerical variables, while categorical variables were imputed and one-hot encoded. This produced 287 transformed columns with no missing or infinite values.

Three independent selection signals were then compared. L1 logistic regression retained 235 non-zero coefficients. LightGBM gain importance used 228 transformed features, while permutation importance identified 55 features with negative contribution on its validation slice. Agreement was strongest between the two tree-based views: LightGBM gain and permutation importance had a Spearman correlation of 0.677 and shared 12 of their top 15 features. Five transformed variables appeared in the top 15 of all three methods, including EXT_SOURCE_2, EXT_SOURCE_3 and AMT_GOODS_PRICE.

Rather than trusting one importance method, the union of the three top-15 lists was mapped back to 21 original source variables: 17 numerical and four categorical. The reduced representation produced 98 transformed columns. In a three-fold LightGBM check, ROC-AUC moved from 0.7702 on the full matrix to 0.7600 on the reduced matrix, while fitting time fell from 49.1 to 15.8 seconds. The reduction therefore accepted a modest loss in ranking performance in exchange for a much smaller and faster working representation.

The final preprocessing pipeline was fitted inside each modelling fold. Three strongly right-skewed non-negative variables were log-transformed, numerical variables were median-imputed and standardised, and categorical variables were imputed before one-hot encoding. This common pipeline ensured that all later model families were compared on the same information under the same leakage-safe five-fold stratified cross-validation design.

The three importance views are shown together in Figure 3. Their rankings are not identical because a sparse linear model, a boosting model and permutation importance measure usefulness differently. The overlap around the external-source variables is therefore more convincing than a high ranking produced by one technique alone.

Disagreement was not treated as an error; it indicated that some variables may contribute through nonlinear effects or interactions that are not captured equally by every method. The union-based strategy retained this complementary information while reducing dimensionality.

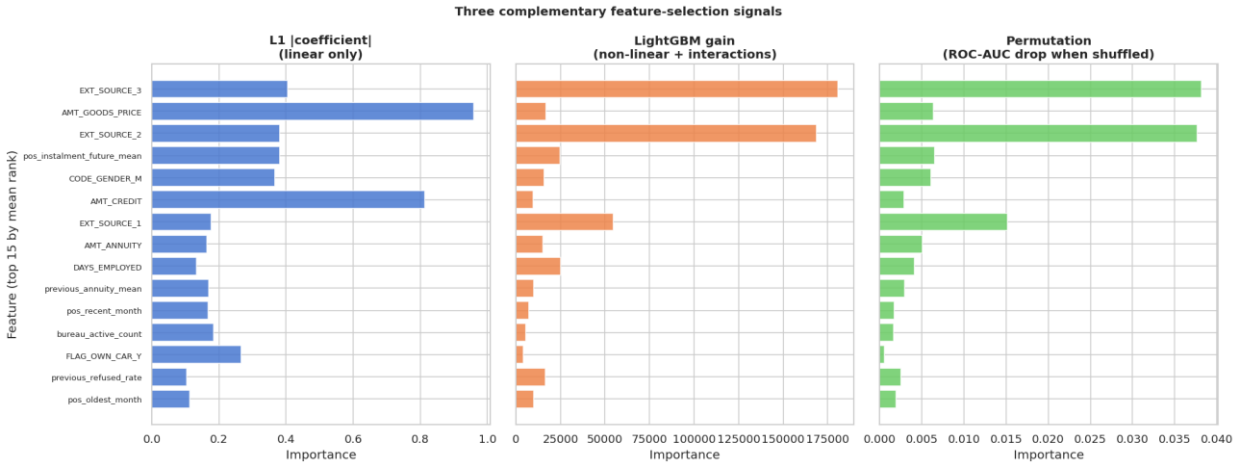


Figure 3. Three complementary feature-selection signals: L1 coefficients, LightGBM gain importance and permutation importance.

Source: executed output from the submitted Home Credit assessment notebook.

4. Model Development

The classical stage established interpretable baselines before moving to ensembles. Logistic regression and linear SVM were the strongest classical models, both reaching about 0.7466 ROC-AUC after tuning. k-NN improved materially when the number of neighbours increased, rising from 0.6945 to 0.7263, but it remained below the linear models. Naive Bayes was weakest at 0.6384. A small RBF SVM probe produced only a 0.0053 ROC-AUC advantage over the linear version while taking about 27 times longer, so the linear SVM was retained for efficiency.

Tree ensembles improved ranking quality. The tuned Random Forest reached 0.7487 ROC-AUC, XGBoost reached 0.7618 and LightGBM reached 0.7621. The Optuna searches were intentionally bounded to 15 trials per ensemble. The small gains for XGBoost and LightGBM showed that extensive tuning was not the main source of performance; the boosting model family itself was the stronger fit for this structured tabular problem.

A matched multilayer perceptron was implemented in both Keras and PyTorch. The same architecture, class weighting, early stopping logic and preprocessing were used so that the framework comparison remained meaningful. A small Keras tuning search selected a learning rate of 0.001, width 64 and dropout 0.2. Five-fold validation then gave mean ROC-AUC of 0.7523 for both Keras and PyTorch, with a mean absolute fold difference of only 0.0006. The paired Wilcoxon result was $p = 1.0000$. This does not prove equivalence, but it shows that the neural result was highly consistent across the two implementations. The neural network remained competitive without overtaking boosting.

The cross-validation plot in Figure 4 shows that the leading boosting models stayed near the top across the shared folds rather than winning because of one favourable split. Their advantage over the neural and classical models was modest but persistent. The result is therefore better described as a stable edge than a dramatic separation. The neural models also performed credibly, suggesting that the tabular representation contained learnable structure across model families. Their additional implementation complexity, however, did not provide enough predictive benefit to replace the boosting choice.

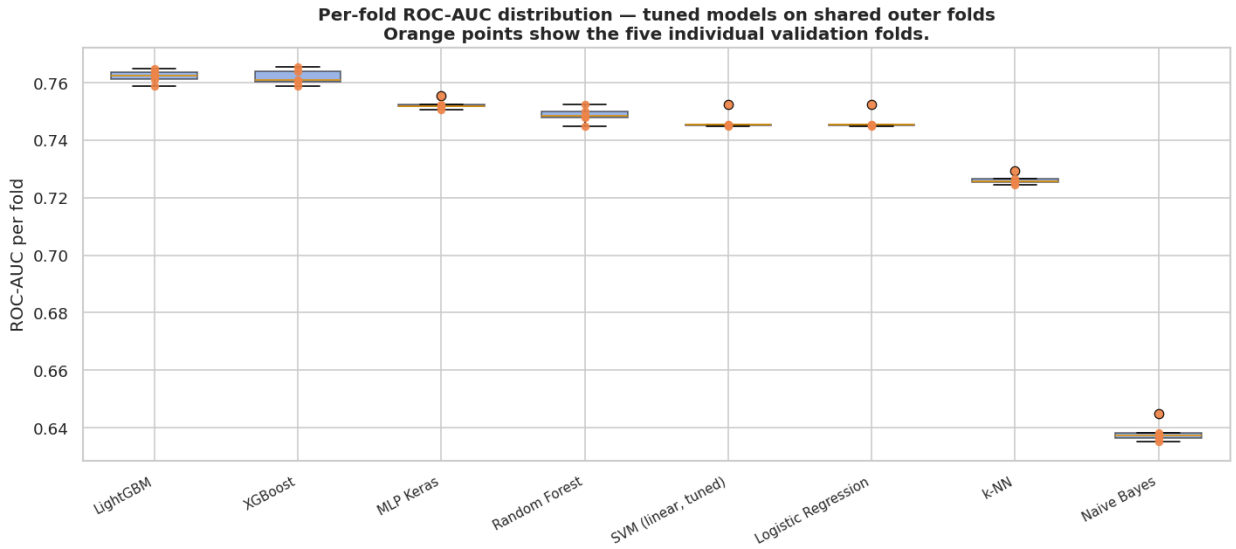


Figure 4. Per-fold ROC-AUC for tuned models on the shared outer folds. LightGBM and XGBoost remain closely matched across validation folds.

Source: executed output from the submitted Home Credit assessment notebook.

5. Comparative Evaluation and Model Selection

The unified development leaderboard placed LightGBM first at 0.7621 ROC-AUC and 0.2485 PR-AUC. XGBoost was almost identical at 0.7618 ROC-AUC and 0.2475 PR-AUC. Keras followed at 0.7523, Random Forest at 0.7487, and logistic regression and linear SVM at 0.7466. The leading PR-AUC values were more than three times the no-skill baseline of 0.0807, showing useful separation of the minority default class.

Recall at the default 0.50 threshold varied sharply between models. This was especially visible for the SVM, which had strong ranking performance but recall of only 0.0114 at 0.50. That result reinforced a key point: ROC-AUC evaluates ranking over thresholds, whereas recall depends on the chosen operating point. Threshold selection therefore had to be handled as a separate business decision rather than inferred from AUC.

A Friedman test across eight model families and five shared folds produced chi-squared = 33.0 with $p = 0.00003$, indicating that the model rankings were not all drawn from the same performance distribution. LightGBM had the best mean rank at 1.40 and XGBoost was next at 1.60. Pairwise Wilcoxon tests were much less informative. With only five paired folds, the smallest attainable exact two-sided p-value was 0.0625, while Bonferroni correction across 28 comparisons required $p < 0.00179$. No pair could meet that threshold. The appropriate conclusion is therefore that the overall model set differs, while individual pairwise claims are limited by the small number of folds. Effect size, direction and consistency across folds are more useful than treating non-significance as evidence of equality.

6. Calibration, Fairness and Cost-Sensitive Decision Rule

Raw probabilities from the class-weighted models were poorly calibrated. LightGBM's out-of-fold mean predicted default probability was 0.3951 even though the actual default rate was 0.0807, giving ECE of 0.3143. Calibration was therefore essential before probabilities were used for decision making.

Platt scaling and isotonic regression were compared using cross-fitted development predictions. Brier score was the primary selection criterion, with ECE used as a supporting diagnostic; this procedure selected Platt scaling without using the held-out labels. On the held-out test set, Platt scaling reduced LightGBM ECE from 0.3167 to 0.0031 and Brier score from 0.1875 to 0.0671. Mean predicted risk became 0.0814, very close to the observed test default rate of 0.0807, while ROC-AUC remained 0.7673. Calibration therefore corrected probability meaning without sacrificing ranking.

Subgroup results were examined by gender, age, education and family status. Gender discrimination was very similar, with an AUC range of only 0.0006 between the two main groups. The age range was wider at

0.0392, with the 60-70 group recording AUC of 0.7293 compared with 0.7685 for ages 30-40. Education showed a much smaller AUC range of 0.0034 in the executed results, while family status varied by 0.0295. These differences do not establish unfairness by themselves, but they identify areas that require monitoring alongside selection rates, true-positive rates, false-positive rates and group sample sizes.

SMOTENC was tested inside the cross-validation folds so that synthetic examples could not leak into validation data. It did not improve the system. ROC-AUC fell to 0.7264 from 0.7621 for class-weighted LightGBM, PR-AUC fell to 0.1964 from 0.2485, and runtime increased from about 33 to 157 seconds. The method was therefore rejected.

The final operating threshold was chosen from calibrated out-of-fold training probabilities using an explicit analytical assumption that a false negative costs five times a false positive. The empirical optimum was 0.1613, close to the theoretical calibrated threshold of 0.1667. When this fixed threshold was applied to the test set, recall was 0.426, precision was 0.263 and the approval rate was 86.9%. Total test cost was 20,186 units, compared with 24,463 at the default 0.50 threshold, a reduction of 4,277 cost units. The 5:1 cost ratio is not supplied by the dataset and must be validated against real lending economics before live use.

Figure 5 illustrates the most important change between raw and calibrated probabilities. Before calibration, class-weighted LightGBM scores were systematically too high relative to the observed default frequency. After Platt scaling, the reliability curve moved closer to the ideal diagonal and average predicted risk aligned with the actual event rate. Figure 6 shows why the final threshold was not set at 0.50. Under the stated cost assumption, minimum expected cost occurred near 0.16. That operating point detected more potential defaulters, but it also reduced approvals and created more false positives. The threshold is therefore a policy trade-off supported by model evidence, not a universal statistical constant.

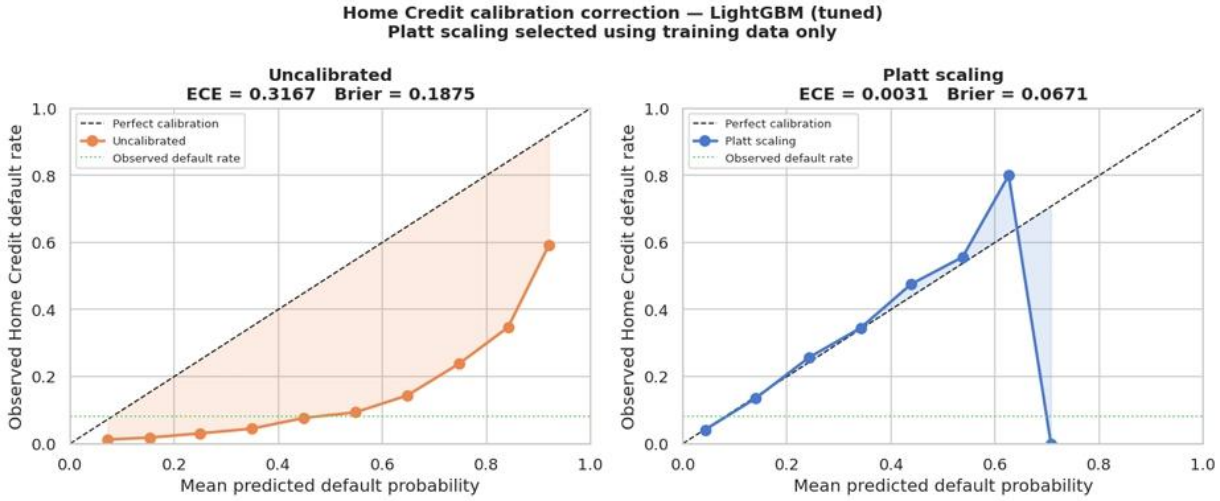


Figure 5. LightGBM calibration correction. Platt scaling moves predicted probabilities much closer to observed default frequencies.

Source: executed output from the submitted Home Credit assessment notebook.

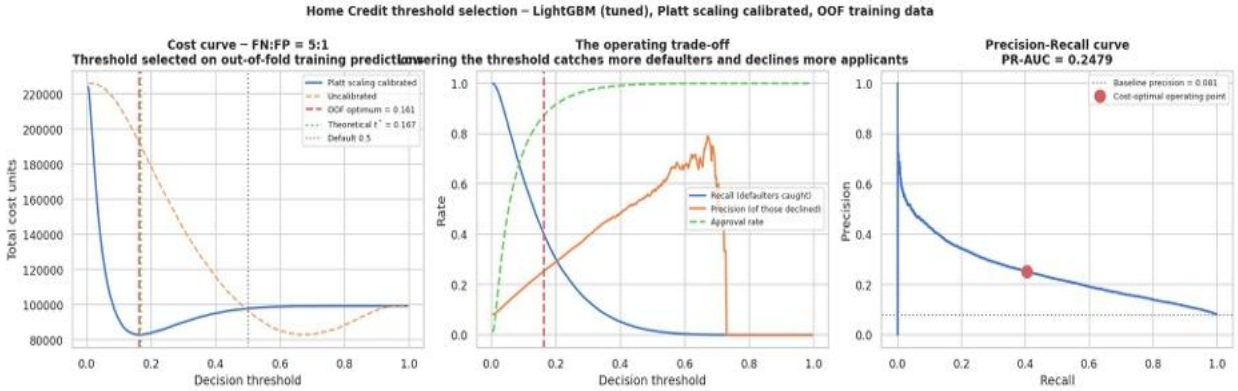


Figure 6. Cost-sensitive threshold analysis. The out-of-fold minimum occurs close to 0.16 under the stated 5:1 false-negative to false-positive cost assumption.

Source: executed output from the submitted Home Credit assessment notebook.

7. Explainability and Deployment Readiness

SHAP analysis was performed on the strongest classical, ensemble and neural models using the same 120 held-out applicants and 98 transformed features. The global rankings were broadly consistent across model families. Spearman agreement was 0.853 between SVM and LightGBM, 0.899 between SVM and Keras, and 0.884 between LightGBM and Keras. EXT_SOURCE_2 and EXT_SOURCE_3 were the two most consistently influential variables, followed by features such as AMT_GOODS_PRICE, EXT_SOURCE_1, AMT_CREDIT, previous repayment information and POS/CASH history.

The local explanation deliberately examined a false positive near the operating boundary. Its calibrated default probability was 0.1664 against a threshold of 0.1613, leaving a margin of only 0.0051. POS instalments remaining was the strongest local risk-increasing contribution. This type of explanation is useful for understanding the model's evidence, but the SHAP values are not causal statements and should not be presented as guarantees that changing one applicant characteristic would change the real outcome.

Held-out evaluation confirmed the development choice. LightGBM achieved test ROC-AUC of 0.7673 and PR-AUC of 0.2603, with prediction latency of about 4.46 ms per 1,000 applicants. XGBoost was extremely close at 0.7667 ROC-AUC and was slightly faster at 3.95 ms per 1,000. The small gap does not justify reopening model selection after observing the test labels. LightGBM remained the pre-test development winner and had already been calibrated and linked to the selected threshold.

The full preprocessing pipeline, fitted LightGBM model, calibrator and threshold were saved and reloaded successfully, showing that the notebook produced a reusable scoring bundle rather than an in-memory experiment. The accompanying model card records intended use, metrics, subgroup results, limitations and monitoring conditions.

The SHAP results in Figure 7 add interpretation to the performance comparison. External credit-source variables were influential across the strongest model families, while application amounts and previous repayment behaviour also appeared repeatedly. This consistency suggests that the models use similar broad signals despite different mathematical structures. Figure 8 places predictive quality beside inference latency. LightGBM and XGBoost occupy a strong region of that trade-off, combining high held-out ROC-AUC with fast scoring. The plot supports the final selection without implying that latency alone should decide the model; calibration, explainability, governance and stability remain important.

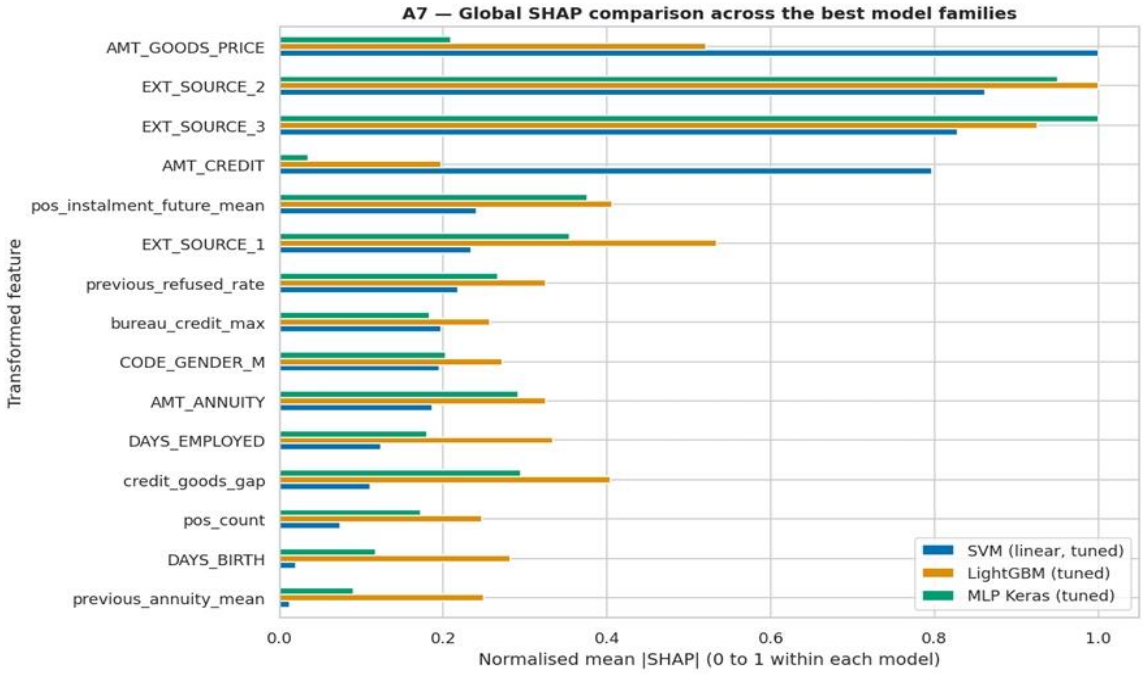


Figure 7. Global SHAP comparison across the strongest model families. External-source variables are consistently among the most influential features.

Source: executed output from the submitted Home Credit assessment notebook.

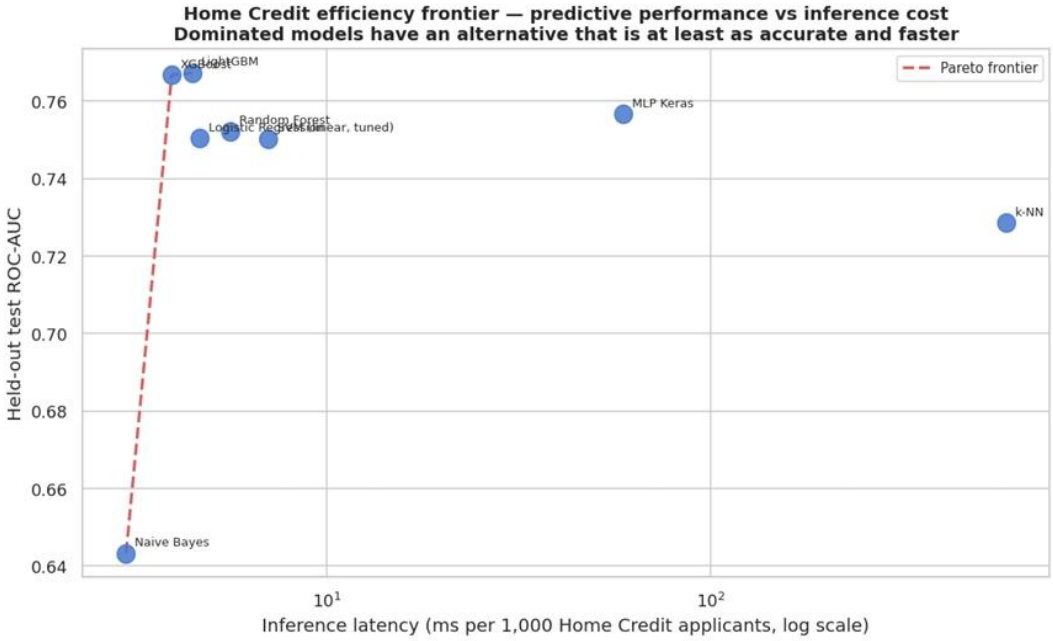


Figure 8. Held-out predictive performance against inference cost. The frontier shows the practical trade-off between ROC-AUC and scoring latency.

Source: executed output from the submitted Home Credit assessment notebook.

8. Limitations and Conclusion

The strongest limitation is structural rather than algorithmic. The model learns from historical lending outcomes and cannot observe what would have happened to applicants who were not financed. Its labels therefore reflect both borrower behaviour and previous lending policy. More data from the same historical process cannot recover those missing counterfactual outcomes.

There are also practical limitations. Five outer folds restrict pairwise statistical power, the chosen threshold depends on an assumed cost ratio, raw class weighting distorted probability calibration, and subgroup behaviour may change as applicant populations and economic conditions change. Performance, calibration and data distributions can all drift after deployment.

The evidence supports a controlled pilot or further validation, not fully automatic lending decisions. The recommended system is a Platt-calibrated, tuned LightGBM model operating at the 0.1613 cost-sensitive threshold, with human review and an appeal route for consequential cases. Its main strength is not a dramatic AUC advantage over XGBoost, but the combination of strong ranking, corrected probabilities, documented threshold logic, explainability and practical latency. Deployment should version the model, preprocessing, calibrator and threshold together, monitor ROC-AUC, PR-AUC, ECE/Brier and subgroup metrics, and require a formal validation step before retraining or recalibration.

Deployment quality ultimately depends on the complete decision process rather than model discrimination considered in isolation alone.

***Source note**: all numerical results and figures in this report are taken from the outputs executed of the submitted Home Credit assessment notebook.*

APENDIX

AI ETHICS DECLARATION:

STATEMENT 2: AI is used for shaping parts of the activity; and/or improving structure/quality of the final output.